

HARSHATH MUKUNDAN

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SUMMARY

Final-year Computer Science (AI & ML) student with hands-on experience building Machine Learning, Deep Learning, and Generative AI systems using Python and SQL. Skilled in developing computer vision, spatiotemporal forecasting, fraud detection, and hybrid RAG solutions, with experience in model development, API integration, MLOps, and cloud deployment. Strong ability to translate complex data problems into scalable, deployable solutions and deliver projects end-to-end from model development to application deployment.

EDUCATION

B.Tech — Computer Science and Engineering (AI & ML)

2023 – 2027

SRM Institute of Science and Technology, Chennai

CGPA: 8.80 / 10 (till 6th Semester) / No Active Backlogs

EXPERIENCE

Document Processing Automation Intern

05 Dec 2025 – 02 Mar 2026

iBrits Business Solutions, Hosur

- Built an OCR + OpenCV pipeline to automate extraction of structured data (e.g. invoices, forms, IDs) from scanned documents, replacing manual data entry for the team.
- Validated extraction outputs against source documents across varied formats and layouts, identifying and fixing preprocessing failure cases (skew, noise, low-contrast scans) to improve pipeline robustness.

Supply Chain Digitalization Intern

02 Dec 2024 – 02 Mar 2025

AAD Technology, Bengaluru (Remote)

- Tested and validated data-exchange integrity for a cloud-based NGAVS integration across MIPO, ASN, GRN, and invoice workflows spanning multiple upstream/downstream systems.
- Traced and documented root causes of data-sync discrepancies in the integration pipeline, producing a supply-chain data-flow reference used by the team

PROJECTS

Fishing Zone Recommendation System (GitHub)

Python · PyTorch · U-Net · ConvLSTM · Flask · Leaflet.js | 2025

- Designed a hybrid U-Net + ConvLSTM architecture for spatial segmentation and 7-day spatio-temporal forecasting of fishing zones using real CMEMS satellite data (SST, SSH, Chlorophyll-a) across 2,914 satellite frames (2018–2024), achieving 92.13% segmentation accuracy ($F1 = 0.902$, $mIoU = 82.29\%$) and 88.49% forecasting accuracy.
- Deployed a Flask web app with a Leaflet.js map dashboard, Chart.js analytics, and GPS-coordinate CSV export; integrated Power BI-compatible exports and SQL-backed session logging.

Smart Invoice Validation & Fraud Detection Platform (GitHub)

Python · Django · MySQL · Jenkins · Docker | 2025

- Co-built a cloud-based GST invoice fraud detection system using Isolation Forest for anomaly detection ($F1 = 0.89$) and Random Forest for behavioral pattern fraud detection ($F1 = 0.91$, $ROC-AUC = 0.97$), validated against a labeled dataset; exposed a real-time fraud-scoring REST API averaging 1.4s response time under 100 concurrent users (JMeter load test).
- Implemented an MLOps pipeline with MLflow experiment tracking and Jenkins CI/CD, containerized with Docker and deployed the application on AWS EC2 behind a load balancer, validated for concurrent request handling.

AI Smart Surveillance & Face Recognition (GitHub)

Python · OpenCV · dlib (ResNet) · SQLite · Docker · Flask | 2024

- Built a real-time face detection and recognition pipeline using dlib's ResNet-based face-recognition model on live CCTV feeds, achieving sub-200ms per-frame inference, automating repetitive monitoring and incident logging that was previously done manually.
- Designed a SQL-backed analytics dashboard with detection-frequency heatmaps and automated incident summary reports, backed by a centralized SQLite event log for audit and forensic use.

FinGraph — AI Financial Document Analyst (GitHub)

React · FastAPI · LangChain · LangGraph · Google Gemini ·

ChromaDB · Neo4j · PostgreSQL | 2026

- Built a hybrid RAG system combining ChromaDB semantic vector search with Neo4j knowledge-graph traversal for multi-hop question answering over SEC financial filings, orchestrated with LangChain/LangGraph and Google Gemini for generation.
- Designed a deterministic Python financial-calculator engine to compute margins and growth rates before LLM generation, reducing numerical errors in financial queries.

TECHNICAL SKILLS

- Programming Languages:** Python, SQL (Advanced), Java, C++, JavaScript
- CS Fundamentals:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks
- Machine Learning & Deep Learning:** Supervised Learning, Regression, Classification, Anomaly Detection, CNN, RNN, U-Net, ConvLSTM, PyTorch, Scikit-learn, EDA
- Generative AI:** LangChain, LangGraph, RAG (Hybrid Vector + Graph), LLM, SentenceTransformers
- Frameworks & Libraries:** OpenCV, Pandas, NumPy, Django, Flask, React, Node.js, Express.js
- Databases & Analytics:** MongoDB, SQLite, PostgreSQL, Neo4j, ChromaDB, Power BI, SAP Analytics Cloud, MS Excel
- Tools & DevOps:** AWS (EC2, S3), Docker, Jenkins, CI/CD, MLflow, Git, GitHub, Linux

CERTIFICATIONS

- SAP Certified – Generative AI Developer | SAP Certified – Data Analyst, SAP Analytics Cloud
- NPTEL – Programming in Java | NPTEL – Introduction to Machine Learning
- MongoDB – Associate Developer | Coursera – Supervised ML: Regression & Classification